

Multiple choice Questions

Assignment - I Linear convolution

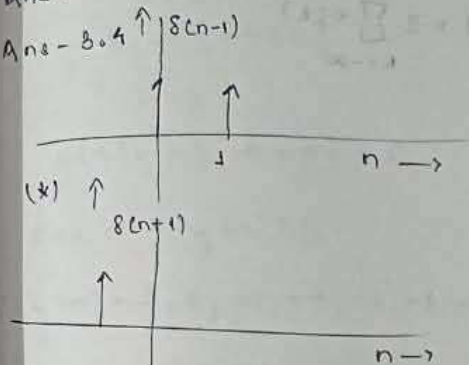
Name: Srikantha Mohanty
(2413068)

Ans-8.1 (b)

Ans-8.2 (b)

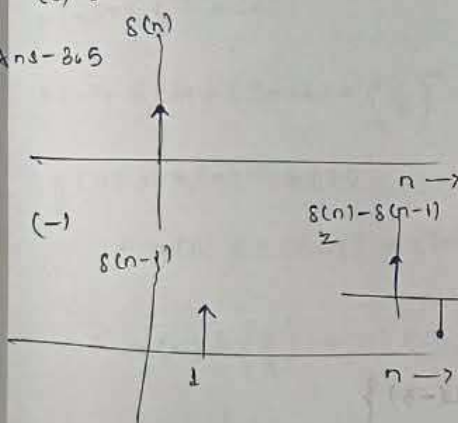
Ans-8.3 (b)

Ans-8.4



(c) $\delta(n)$

Ans-8.5



(b)

Ans-8.6 (c)

Ans-8.7 (c)

Ans-8.8 (a)

Ans-8.9 (c)

Ans-8.10 (c)

$$\text{Ans-8.11 } h(n) = a^n u(n)$$

(d)

Ans-8.12

$$a^n u(n), 0 < a < 1$$

(b)

Ans-8.13

$$y(n) = u(n) * u(n-1)$$

$$y(5) = \sum_{k=-\infty}^{\infty} u(k) u(5-k-1)$$

$$= \sum_{k=0}^{\infty} u(k) u(4-k)$$

$$= \sum_{k=0}^4 u(k) u(4-k)$$

$$= u(0)u(4) + u(1)u(3) + u(2)u(2) + u(3)u(1) + u(4)u(0)$$

$$= 2 \quad (b)$$

Ans-8.14

(d)

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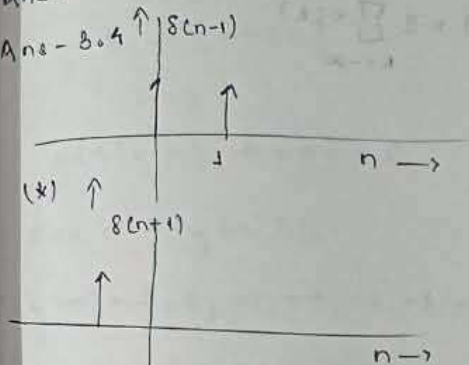
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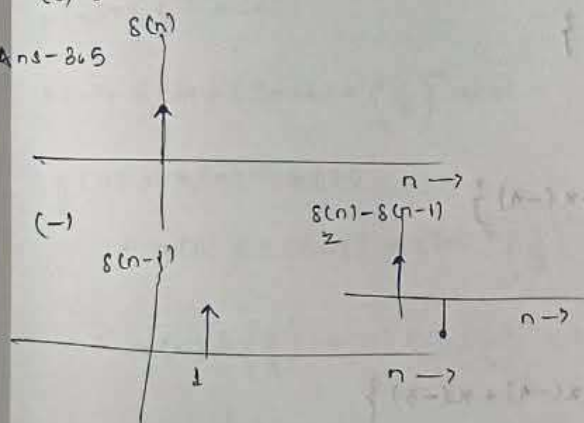
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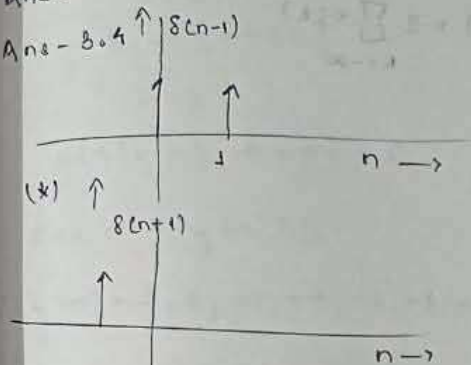
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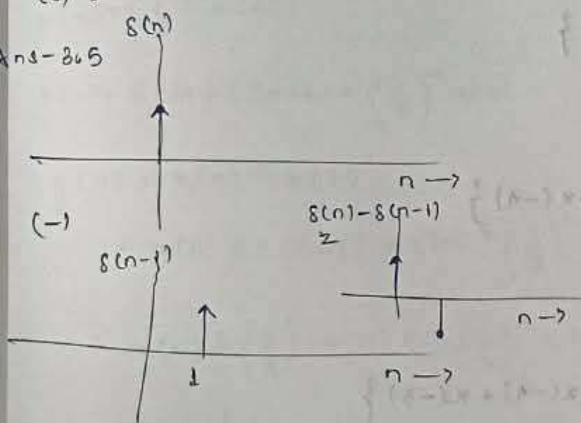
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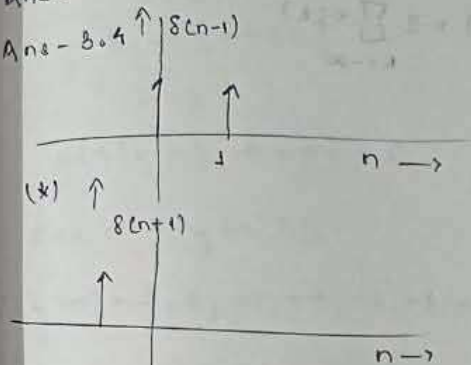
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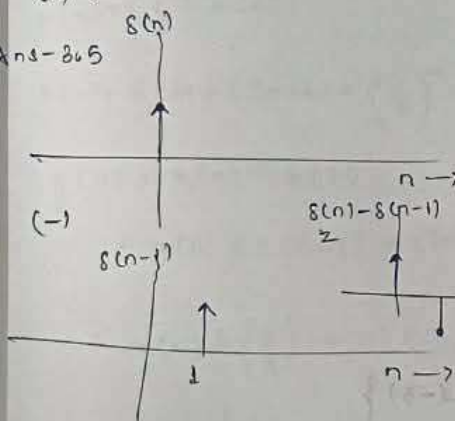
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Problem 3:

Ans - 8.1 (a)

$$x(n) = \begin{cases} -1 & -6 \leq n \leq -1 \\ 1 & 0 \leq n \leq 4 \end{cases}$$

$$h(n) = 2u(n)$$

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k) = 2 \sum_{k=-\infty}^{\infty} x(k)u(n-k) = 2 \sum_{k=-\infty}^n x(k)$$

~~For $n < -6$, $y(n) = 0$~~

~~For $n = -6$~~

~~$y(-6) = 2 \sum_{k=-\infty}^{-6} x(k)$~~

For $n < -6$, $y(n) = 0$

$$y(-6) = 2 \sum_{k=-\infty}^{-6} x(k)$$

$$= 2 \{ \dots + x(-6) + x(-5) \}$$

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Similarly, $y(-1) = 2 \{ \dots + x(-5) + x(-1) \}$

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$$y(-3) = 2 \{ \dots + x(-5) + x(-1) + x(-3) \}$$

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$$y(-2) = 2 \{ \dots + x(-5) + x(-1) + x(-3) + x(-2) \}$$

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$$y(-1) = 2 \{ -1-1-1-1-1 \}$$

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For $n > 4$ $y(n) = 0$

$$\therefore y(n) = \{0, -2, -4, -6, -8, -10, -8, -6, -4, -2, 0, \dots\}$$

Ans - 8.1 (b)

$$x(n) = \left(\frac{1}{2}\right)^n u(n)$$

$$h(n) = 8(n) + 8(n-1) + \left(\frac{1}{3}\right)^n u(n)$$

$$y(n) = x(n) * h(n)$$

$$= x(n) + x(n-1) + x(n) * \left(\frac{1}{3}\right)^n u(n)$$

$$x(n) * \left(\frac{1}{3}\right)^n u(n) = 3\left(\frac{1}{2}\right)^n - 2\left(\frac{1}{3}\right)^n$$

$$\therefore y(n) = \begin{cases} 0 & , n < 0 \\ 2 & , n = 0 \\ 6\left(\frac{1}{2}\right)^n - 2\left(\frac{1}{3}\right)^n & , n \geq 1 \end{cases}$$

Ans - 8.1 (c)

$$x(n) = u(n)$$

$$h(n) = 1, 0 \leq n \leq 9$$

$$y(n) = x(n) * h(n)$$

$$y(0) = 2\{-1-1-1-1-1+1\}$$

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$$x(n) = u(n)$$

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$$z = \begin{cases} 0, & n < 0 \\ n+1, & 0 \leq n \leq 9 \\ 10, & n \geq 9 \end{cases}$$

Ans-3.1(d)

$$x(n) = \left(\frac{1}{3}\right)^n u(n)$$

$$h(n) = \delta(n) + \left(\frac{1}{3}\right)^n u(n)$$

$$y(n) = x(n) * h(n)$$

$$= x(n) * \left(\delta(n) + \left(\frac{1}{3}\right)^n u(n) \right)$$

$$= x(n) + \left(\frac{1}{3}\right)^n u(n) (n+1)$$

$$= (n+2) \left(\frac{1}{3}\right)^n u(n)$$

Ans-3.2 $h_1(n) = \left(\frac{1}{2}\right)^n u(n)$

$$h_2(n) = \left(\frac{1}{4}\right)^n u(n)$$

$$h(n) = h_1(n) * h_2(n)$$

$$= \sum_{k=0}^n \left(\frac{1}{2}\right)^k \left(\frac{1}{4}\right)^{n-k}$$

$$= \left[2 \left(\frac{1}{2}\right)^n - \left(\frac{1}{4}\right)^n \right] u(n)$$

Ans-3.3 $h(n) = \alpha^n u(n)$

For BIBO stability

$$\sum_{n=-\infty}^{\infty} |h(n)| < \infty$$

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$$= 2 \{ -1 \} = -8$$

$$y(-1) = 2 \{ -1-1-1-1-1 \}$$

$$= -10$$

$$\Rightarrow \sum_{n=0}^{\infty} |\alpha|^n < \infty$$

$$\Rightarrow \frac{1}{1-|\alpha|} < \frac{1}{0}$$

$$\Rightarrow 1-|\alpha| > 0$$

$$\Rightarrow |\alpha| < 1$$

Ans-8.4

$$h(n) = \begin{cases} \alpha^n, & n \geq 0 \\ \beta^n, & n < 0 \end{cases}$$

For $n \geq 0$

$$|\alpha| < 1$$

For $n < 0$, as $n \rightarrow -\infty$

$$\sum_{n=-\infty}^0 |\beta|^n < \infty$$

$$\Rightarrow \frac{-1}{1-|\beta|} < \frac{1}{0}$$

$$\Rightarrow |\beta| - 1 > 0$$

$$\Rightarrow |\beta| > 1$$

Ans-8.5 $x(n) = a^n u(n)$, $0 < a < 1$

$$\begin{aligned} R_{xx}(n) &= \sum_m x(n) x(n-m) \\ &= \frac{a^{|n|}}{1-a^2} \end{aligned}$$

Ans-8.6(a) $h(n) = \left(\frac{1}{2}\right)^n u(n)$

$$u(n) = 0, n < 0$$

$$h(n) = 0, n < 0$$

System is causal

$$\sum_{n=-\infty}^{\infty} |h(n)|$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n$$

$$= \frac{1}{1-\frac{1}{2}} = 2 < \infty$$

$\Rightarrow$ System is stable

Ans-8.4(b)

$$h(n) = \left(\frac{1}{2}\right)^n u(n-1)$$

$$h(n) = 0 \text{ for } n < 1$$

$\Rightarrow$ System is causal

$$\sum_{n=-\infty}^{\infty} |h(n)|$$

$$= \sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n$$

$$= \frac{\frac{1}{2}}{1-\frac{1}{2}} = 1 < \infty$$

$\Rightarrow$ System is stable

Ans-8.6(c)

$$h(n) = \left(\frac{1}{2}\right)^{|n|}$$

$$n = -1$$

$$h(-1) = \left(\frac{1}{2}\right)^{|-1|} = \frac{1}{2}$$

$$h(-1) \neq 0$$

$\Rightarrow$ Non-causal

$$\Rightarrow \sum_{n=0}^{\infty} |\alpha|^n < \infty$$

$$\Rightarrow \frac{1}{1-|\alpha|} < \frac{1}{0}$$

$$\Rightarrow 1-|\alpha| > 0$$

$$\Rightarrow |\alpha| < 1$$

Ans-8.4

$$h(n) = \begin{cases} \alpha^n, & n \geq 0 \\ \beta^n, & n < 0 \end{cases}$$

For $n \geq 0$

$$|\alpha| < 1$$

For $n < 0$, as $n \rightarrow -\infty$

$$\sum_{n=-\infty}^0 |\beta|^n < \infty$$

$$\Rightarrow \frac{-1}{1-|\beta|} < \frac{1}{0}$$

$$\Rightarrow |\beta| - 1 > 0$$

$$\Rightarrow |\beta| > 1$$

Ans-8.5 $x(n) = a^n u(n)$, $0 < a < 1$

$$\begin{aligned} R_{xx}(n) &= \sum_m x(n) x(n-m) \\ &= \frac{a^{|n|}}{1-a^2} \end{aligned}$$

Ans-8.6(a) $h(n) = \left(\frac{1}{2}\right)^n u(n)$

$$u(n) = 0, n < 0$$

$$h(n) = 0, n < 0$$

System is causal

$$\sum_{n=-\infty}^{\infty} |h(n)|$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n$$

$$= \frac{1}{1-\frac{1}{2}} = 2 < \infty$$

$\Rightarrow$ System is stable

Ans-8.4(b)

$$h(n) = \left(\frac{1}{2}\right)^n u(n-1)$$

$$h(n) = 0 \text{ for } n < 1$$

$\Rightarrow$ System is causal

$$\sum_{n=-\infty}^{\infty} |h(n)|$$

$$= \sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n$$

$$= \frac{\frac{1}{2}}{1-\frac{1}{2}} = 1 < \infty$$

$\Rightarrow$ System is stable

Ans-8.6(c)

$$h(n) = \left(\frac{1}{2}\right)^{|n|}$$

$$n = -1$$

$$h(-1) = \left(\frac{1}{2}\right)^{|-1|} = \frac{1}{2}$$

$$h(-1) \neq 0$$

$\Rightarrow$ Non-causal

$$\Rightarrow \sum_{n=0}^{\infty} |\alpha|^n < \infty$$

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Ans-8.4

$$h(n) = \begin{cases} \alpha^n, & n \geq 0 \\ \beta^n, & n < 0 \end{cases}$$

For $n \geq 0$

$$|\alpha| < 1$$

For $n < 0$, as $n \rightarrow -\infty$

$$\sum_{n=-\infty}^0 |\beta|^n < \infty$$

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Ans-8.5 $x(n) = a^n u(n)$, $0 < a < 1$

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$$\sum_{n=-\infty}^{\infty} |h(n)|$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n$$

$$= \frac{1}{1-\frac{1}{2}} = 2 < \infty$$

$\Rightarrow$ System is stable

Ans-8.4(b)

$$h(n) = \left(\frac{1}{2}\right)^n u(n-1)$$

$$h(n) = 0 \text{ for } n < 1$$

$\Rightarrow$ System is causal

$$\sum_{n=-\infty}^{\infty} |h(n)|$$

$$= \sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n$$

$$= \frac{\frac{1}{2}}{1-\frac{1}{2}} = 1 < \infty$$

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Ans-8.6(c)

$$h(n) = \left(\frac{1}{2}\right)^{|n|}$$

$$n = -1$$

$$h(-1) = \left(\frac{1}{2}\right)^{|-1|} = \frac{1}{2}$$

$$h(-1) \neq 0$$

$\Rightarrow$ Non-causal

$$\sum_{n=-\infty}^{\infty} \left(\frac{1}{2}\right)^{|n|}$$

$$= 1 + 2 \sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n$$

$$= 1 + 2(1) = 3 < \infty$$

∴ System is stable

Ans → 3.6 (d)

$$h(n) = u(n+2) - u(n-2) \quad h(n) = \begin{cases} 1, & -2 \leq n \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

$$u(n+2) = 1 \text{ for } n \geq -2$$

$$h(-2) = 1$$

⇒ System is non-causal

$$\sum_{n=-\infty}^{\infty} |h(n)|$$

$$= 1 + 1 + 1 + 1 = 4 < \infty$$

∴ System is stable.

Ans → 3.7 $y(n) = ny(n-1) + x(n)$

$$y(-1) = 0$$

For impulse response i

$$x(n] = \delta(n)$$

$$y(0) = 0y(-1) + x(0) \\ = 1$$

$$y(1) = 1y(0) + x(1) \\ = 1$$

$$y(2) = 2y(1) + x(2) \\ = 2 \dots$$

Hence, not LTI and not BIBO stable

$$\sum_{n=-\infty}^{\infty} \left(\frac{1}{2}\right)^{|n|}$$

$$= 1 + 2 \sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n$$

$$= 1 + 2(1) = 3 < \infty$$

∴ system is stable

Ans → 3.6 (d)

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Hence, not LTI and not BIBO stable

Ans - 8.8

$$(a) x = \left\{ 1, -\frac{1}{2}, \frac{1}{4}, -\frac{1}{8}, \frac{1}{16} \right\}$$

$$h = \left\{ 1, -1, 1, -1 \right\}$$

$$y(n) = x(n) * h(n)$$

$$\text{No. of samples} = 5 + 4 - 1 = 8$$

$$y(0) = 1(1) = 1$$

$$y(1) = 1(-1) + \left(-\frac{1}{2}\right)(1) = -\frac{3}{2}$$

$$y(2) = 1(1) + \left(-\frac{1}{2}\right)(-1) + \frac{1}{4}(1) = \frac{7}{4}$$

$$y(3) = 1(-1) + \left(-\frac{1}{2}\right)(1) + \frac{1}{4}(-1) + \left(-\frac{1}{8}\right)(1) \\ = -\frac{15}{8}$$

$$y(4) = \left(-\frac{1}{2}\right)(-1) + \frac{1}{4}(1) + \left(-\frac{1}{8}\right)(-1) + \frac{1}{16}(1) \\ = \frac{15}{16}$$

$$y(5) = \frac{1}{4}(-1) + \left(-\frac{1}{8}\right)(1) + \frac{1}{16}(-1) \\ = -\frac{7}{16}$$

$$y(6) = \left(-\frac{1}{8}\right)(-1) + \frac{1}{16}(1) \\ = \frac{1}{8} + \frac{1}{16} = \frac{3}{16}$$

$$y(7) = \frac{1}{16}(-1) \\ = -\frac{1}{16}$$

Ans - 8.8

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Ans - 8.8

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$$\sum_{n=-\infty}^{\infty} \left(\frac{1}{2}\right)^{|n|}$$

$$= 1 + 2 \sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n$$

$$= 1 + 2(1) = 3 < \infty$$

∴ system is stable

Ans → 3.6 (d)

$$h(n) = u(n+2) - u(n-2) \quad h(n) = \begin{cases} 1, & -2 \leq n \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

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⇒ system is non-causal

$$\sum_{n=-\infty}^{\infty} |h(n)|$$

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$$y(7) = \frac{1}{16}(-1) \\ = -\frac{1}{16}$$

$$\Rightarrow \sum_{n=0}^{\infty} |\alpha|^n < \infty$$

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$$\Rightarrow 1-|\alpha| > 0$$

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Ans-8.5 $x(n) = a^n u(n)$, $0 < a < 1$

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Ans-8.6(a) $h(n) = \left(\frac{1}{2}\right)^n u(n)$

$$u(n) = 0, n < 0$$

$$h(n) = 0, n < 0$$

System is causal

$$\sum_{n=-\infty}^{\infty} |h(n)|$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n$$

$$= \frac{1}{1-\frac{1}{2}} = 2 < \infty$$

$\Rightarrow$ System is stable

Ans-8.4(b)

$$h(n) = \left(\frac{1}{2}\right)^n u(n-1)$$

$$h(n) = 0 \text{ for } n < 1$$

$\Rightarrow$ System is causal

$$\sum_{n=-\infty}^{\infty} |h(n)|$$

$$= \sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n$$

$$= \frac{\frac{1}{2}}{1-\frac{1}{2}} = 1 < \infty$$

$\Rightarrow$ System is stable

Ans-8.6(c)

$$h(n) = \left(\frac{1}{2}\right)^{|n|}$$

$$n = -1$$

$$h(-1) = \left(\frac{1}{2}\right)^{|-1|} = \frac{1}{2}$$

$$h(-1) \neq 0$$

$\Rightarrow$ Non-causal

$$\therefore y(n) = \left\{ 1, -\frac{3}{2}, \frac{7}{4}, -\frac{15}{8}, \frac{15}{16}, -\frac{7}{16}, \frac{3}{16}, -\frac{1}{16} \right\}$$

Ans - 8.8(b)

$$x(n) = \{1, 2, 3, 0, -1\}$$

$$h(n) = \{2, -1, 3, 1, -2\}$$

No. of samples: $5+5-1=9$

$$y(0) = 1(2) = 2$$

$$y(1) = 1(-1) + 2(2) = 3$$

$$y(2) = 1(3) + 2(-1) + 3(2) = 7$$

$$y(3) = 1(1) + 2(3) + 3(-1) + 0(2) = 4$$

$$y(4) = 1(-2) + 2(1) + 3(3) + 0(-1) + (-1)(2) = 7$$

$$y(5) = 2(-2) + 3(1) + 0(3) + (-1)(-1) = 0$$

$$y(6) = 3(-2) + 0(1) + (-1)(3) = -9$$

$$y(7) = 0(-2) + (-1)(1) = -1$$

$$y(8) = (-1)(-2) = 2$$

$$\therefore y(n) = \{2, 3, 7, 4, 7, 0, -9, -1, 2\}$$

Ans - 8.8(c)

$$x(n) = \left\{ 3, \frac{1}{2}, -\frac{1}{4}, 1, 1 \right\}$$

$$h(n) = \left\{ 2, -1, \frac{1}{2}, -\frac{1}{2} \right\}$$

Again, no. of samples = $5+4-1=8$

$$y(0) = 3(2) = 6$$

$$y(1) = 3(-1) + \frac{1}{2}(2) = -2$$

$$y(2) = 3\left(\frac{1}{2}\right) + \frac{1}{2}(-1) + \left(-\frac{1}{4}\right)(2)$$

$$= \frac{1}{2}$$

$$y(3) = 3\left(-\frac{1}{2}\right) + \frac{1}{2}\left(\frac{1}{2}\right) + \left(-\frac{1}{4}\right)(-1) + 1(2)$$

$$= 1$$

$$y(3) = \sum_{k=0}^3 x(k) h(3-k)$$

$$= x(0)h(3) + x(1)h(2) + x(2)h(1) + x(3)h(0)$$

$$y(4) = \frac{1}{2} \left(-\frac{1}{2}\right) + \left(-\frac{1}{4}\right) \left(\frac{1}{2}\right) + 1(-1) + 1(2) = \frac{53}{8}$$

$$y(5) = \left(-\frac{1}{4}\right) \left(-\frac{1}{2}\right) + 1 \left(\frac{1}{2}\right) + 1(-1)$$

$$= -\frac{27}{8}$$

$$y(6) = 1 \left(-\frac{1}{2}\right) + 1 \left(\frac{1}{2}\right)$$

$$= \frac{3}{2}$$

$$y(7) = 1 \left(-\frac{1}{2}\right)$$

$$= -2$$

$$\therefore y(n) = \left\{ 0, -2, \frac{1}{2}, 1, \frac{53}{8}, -\frac{27}{8}, \frac{3}{2}, -2 \right\}$$

Ans - 3.9

$$x(n) = 8(n) + 28(n-1) - 8(n-3)$$

$$h(n) = 28(n+1) + 28(n-1)$$

$$(a) y_1(n) = x(n) * h(n)$$

$$\text{using } 8(n-k) * h(n) = h(n-k)$$

$$y_1(n) = h(n) + 2h(n-1) - h(n-3)$$

$$= 28(n+1) + 28(n-1) + 48(n) + 48(n-2) - 28(n-2) - 28(n-4)$$

$$y_1(n) = \{ \dots, 0, -2, 0, 0, 2, 1, 2, 2 \}$$

$$(b) y_2(n) = x(n+2) * h(n)$$

$$x(n+2) = 8(n+2) + 28(n+1) - 8(n-1)$$

$$y_2(n) = h(n+2) + 2h(n+1) - h(n-1)$$

$$= 28(n+3) + 28(n+1) + 48(n+2) + 48(n) - 28(n) - 28(n-2)$$

$$y_2(n) = \{ \dots, 0, 2, 1, 2, 2, 0, -2 \}$$

$$y(4) = \frac{1}{2} \left(-\frac{1}{2}\right) + \left(-\frac{1}{4}\right) \left(\frac{1}{2}\right) + 1(-1) + 1(2) = \frac{53}{8}$$

$$y(5) = \left(-\frac{1}{4}\right) \left(-\frac{1}{2}\right) + 1 \left(\frac{1}{2}\right) + 1(-1)$$

$$= -\frac{27}{8}$$

$$y(6) = 1 \left(-\frac{1}{2}\right) + 1 \left(\frac{1}{2}\right)$$

$$= \frac{3}{2}$$

$$y(7) = 1 \left(-\frac{1}{2}\right)$$

$$= -2$$

$$\therefore y(n) = \left\{ 0, -2, \frac{1}{2}, 1, \frac{53}{8}, -\frac{27}{8}, \frac{3}{2}, -2 \right\}$$

Ans - 3.9

$$x(n) = 8(n) + 28(n-1) - 8(n-3)$$

$$h(n) = 28(n+1) + 28(n-1)$$

$$(a) y_1(n) = x(n) * h(n)$$

$$\text{using } 8(n-k) * h(n) = h(n-k)$$

$$y_1(n) = h(n) + 2h(n-1) - h(n-3)$$

$$= 28(n+1) + 28(n-1) + 48(n) + 48(n-2) - 28(n-2) - 28(n-4)$$

$$y_1(n) = \{ \dots, 0, -2, 0, 0, 2, 1, 2, 2 \}$$

$$(b) y_2(n) = x(n+2) * h(n)$$

$$x(n+2) = 8(n+2) + 28(n+1) - 8(n-1)$$

$$y_2(n) = h(n+2) + 2h(n+1) - h(n-1)$$

$$= 28(n+3) + 28(n+1) + 48(n+2) + 48(n) - 28(n) - 28(n-2)$$

$$y_2(n) = \{ \dots, 0, 2, 1, 2, 2, 0, -2 \}$$

$$(c) y_0(n) = x(n) * h(n+2)$$

$$h(n+2) = 2\delta(n+3) + 2\delta(n+1)$$

$$y_0(n) = 2x(n+3) + 2x(n+1)$$

$$= 2\delta(n+3) + 4\delta(n+2) - 2\delta(n) + 2\delta(n+1) + 4\delta(n) - 2\delta(n-2)$$

$$y_0(n) = \{ \dots, 0, 2, 4, 2, 2, 0, -2 \}$$

Ans - 3.10

$$x(n) = \left(\frac{1}{2}\right)^{n-2} u(n-2)$$

$$h(n) = u(n+2)$$

$$y(n) = x(n) * h(n)$$

$$y(n) = \sum_{k=-\infty}^{\infty} x(k) h(n-k)$$

$$= \sum_{k=-\infty}^{\infty} \left(\frac{1}{2}\right)^{k-2} u(k-2) u(n-k+2)$$

$$= \sum_{k=2}^{n+2} \left(\frac{1}{2}\right)^{k-2}$$

$$m = k-2$$

$$k=2, m=0$$

$$k=n+2, m=n$$

$$= \sum_{m=0}^n \left(\frac{1}{2}\right)^m$$

$$= 1 - \left(\frac{1}{2}\right)^{n+1}$$

$$\frac{1}{2}$$

$$= 2 \left(1 - \left(\frac{1}{2}\right)^{n+1}\right)$$

$$= 2 - 2^{-n}$$

$$y(n) = \{ 1, 1.5, 1.75, 1.875, \dots \}$$

$$y(0) = 2 - 2^0$$

$$= 1$$

$$y(1) = 2 - 2^{-1}$$

$$= \frac{3}{2} = 1.5$$

$$y(2) = 2 - 2^{-2}$$

$$= 2 - \frac{1}{4}$$

$$= \frac{7}{4} = 1.75$$

$$\vdots$$

$$(c) y_0(n) = x(n) * h(n+2)$$

$$h(n+2) = 2\delta(n+3) + 2\delta(n+1)$$

$$y_0(n) = 2x(n+3) + 2x(n+1)$$

$$= 2\delta(n+3) + 4\delta(n+2) - 2\delta(n) + 2\delta(n+1) + 4\delta(n) - 2\delta(n-2)$$

$$y_0(n) = \{ \dots, 0, 2, 4, 2, 2, 0, -2 \}$$

Ans - 3.10

$$x(n) = \left(\frac{1}{2}\right)^{n-2} u(n-2)$$

$$h(n) = u(n+2)$$

$$y(n) = x(n) * h(n)$$

$$y(n) = \sum_{k=-\infty}^{\infty} x(k) h(n-k)$$

$$= \sum_{k=-\infty}^{\infty} \left(\frac{1}{2}\right)^{k-2} u(k-2) u(n-k+2)$$

$$= \sum_{k=2}^{n+2} \left(\frac{1}{2}\right)^{k-2}$$

$$m = k-2$$

$$k=2, m=0$$

$$k=n+2, m=n$$

$$= \sum_{m=0}^n \left(\frac{1}{2}\right)^m$$

$$= 1 - \left(\frac{1}{2}\right)^{n+1}$$

$$\frac{1}{2}$$

$$= 2 \left(1 - \left(\frac{1}{2}\right)^{n+1}\right)$$

$$= 2 - 2^{-n}$$

$$y(n) = \{ 1, 1.5, 1.75, 1.875, \dots \}$$

$$y(0) = 2 - 2^0$$

$$= 1$$

$$y(1) = 2 - 2^{-1}$$

$$= \frac{3}{2} = 1.5$$

$$y(2) = 2 - 2^{-2}$$

$$= 2 - \frac{1}{4}$$

$$= \frac{7}{4} = 1.75$$

$$\vdots$$

$$\therefore y(n) = \left\{ 1, -\frac{3}{2}, \frac{7}{4}, -\frac{15}{8}, \frac{15}{16}, -\frac{7}{16}, \frac{3}{16}, -\frac{1}{16} \right\}$$

Ans - 8.8(b)

$$x(n) = \{1, 2, 3, 0, -1\}$$

$$h(n) = \{2, -1, 3, 1, -2\}$$

No. of samples: $5+5-1=9$

$$y(0) = 1(2) = 2$$

$$y(1) = 1(-1) + 2(2) = 3$$

$$y(2) = 1(3) + 2(-1) + 3(2) = 7$$

$$y(3) = 1(1) + 2(3) + 3(-1) + 0(2) = 4$$

$$y(4) = 1(-2) + 2(1) + 3(3) + 0(-1) + (-1)(2) = 7$$

$$y(5) = 2(-2) + 3(1) + 0(3) + (-1)(-1) = 0$$

$$y(6) = 3(-2) + 0(1) + (-1)(3) = -9$$

$$y(7) = 0(-2) + (-1)(1) = -1$$

$$y(8) = (-1)(-2) = 2$$

$$\therefore y(n) = \{2, 3, 7, 4, 7, 0, -9, -1, 2\}$$

Ans - 8.8(c)

$$x(n) = \left\{ 3, \frac{1}{2}, -\frac{1}{4}, 1, 1 \right\}$$

$$h(n) = \left\{ 2, -1, \frac{1}{2}, -\frac{1}{2} \right\}$$

Again, no. of samples = $5+4-1=8$

$$y(0) = 3(2) = 6$$

$$y(1) = 3(-1) + \frac{1}{2}(2) = -2$$

$$y(2) = 3\left(\frac{1}{2}\right) + \frac{1}{2}(-1) + \left(-\frac{1}{4}\right)(2)$$

$$= \frac{1}{2}$$

$$y(3) = 3\left(-\frac{1}{2}\right) + \frac{1}{2}\left(\frac{1}{2}\right) + \left(-\frac{1}{4}\right)(-1) + 1(2)$$

$$= 1$$

$$y(3) = \sum_{k=0}^3 x(k) h(3-k)$$

$$= x(0)h(3) + x(1)h(2) + x(2)h(1) + x(3)h(0)$$

Ans - 8.12

$$h(n) = (-1)^n (u(n+2) - u(n-3))$$

$$y(n) = h(n) * u(n)$$

$$= \sum_{k=0}^n h(k)$$

$$= \sum_{k=0}^n (-1)^k u(k+2) - (-1)^k u(k-3)$$

$$= \sum_{k=0}^n (-1)^k - \sum_{k=3}^n 1$$

$$= \frac{1 + (-1)^n}{2} - (n-2)$$

$$= \frac{1 + (-1)^n - n + 2}{2}$$

$$= \left\{ 1, 0, \underset{\uparrow}{1}, 0, 1, -1, -1, \dots \right\}$$

Ans - 8.13 (a) $h(n)$ impulse response is 0

except on $N_0 \leq n \leq N_1$ and $N_2 \leq n \leq N_3$

and e/p $x(n)$ is 0

on $N_2 \leq n \leq N_3$

$$y(n) = x(n) * h(n)$$

$$= \sum_{k=-\infty}^{\infty} x(k) h(n-k)$$

$$x(k) \neq 0 \quad N_2 \leq k \leq N_3$$

$$h(n-k) \neq 0 \quad N_0 \leq n-k \leq N_1$$

$$\Rightarrow N_0 + k \leq n \leq N_1 + k$$

$$n_{\min} = N_0 + N_2$$

$$n_{\max} = N_1 + N_3$$

$$\text{Hence, } N_0 + N_2 \leq n \leq N_1 + N_3$$

(b)

$x(n) \neq 0$ N consecutive pts.

$h(n) \neq 0$ M consecutive pts.

No. of o/p points = $N+M-1$

Ans - 8.14

$$h_1(n) = \delta(n) + \frac{1}{2}\delta(n-1)$$

$$h_2(n) = \frac{1}{2}\delta(n) - \frac{1}{4}\delta(n-1)$$

$$h_3(n) = 2\delta(n)$$

upper path

$$x(n) \rightarrow h_1(n)$$

$$h_4(n) = -2\left(\frac{1}{2}\right)^n u(n)$$

$$y_1(n) = x(n) * h_1(n)$$

Lower path

$$x(n) \rightarrow h_2(n)$$

$$h_{04}(n) = h_3(n) + h_4(n)$$

$$h_2(n) * [h_3(n) + h_4(n)]$$

$$h(n) = h_1(n) + h_2(n) * h_3(n) + h_2(n) * h_4(n)$$

$$h_3(n) = 2\delta(n)$$

$$h_4(n) = -2\left(\frac{1}{2}\right)^n u(n)$$

$$h_3(n) + h_4(n) = 2\delta(n) - 2\left(\frac{1}{2}\right)^n u(n)$$

$$h_2(n) * 2\delta(n) = 2h_2(n)$$

$$2h_2(n) = \delta(n) - \frac{1}{2}\delta(n-1)$$

$$h_2(n) * h_4(n) = \frac{1}{2}h_4(n) - \frac{1}{4}h_4(n-1) = -\left(\frac{1}{2}\right)^n u(n) + \left(\frac{1}{2}\right)^n u(n-1)$$

$$z = -\delta(n)$$

Ans - 8.11

$$h_1(n) * h_2(n)$$

$$h_1(n) = h_2(n) = \left(\frac{1}{3}\right)^n u(n)$$

$$h_1(n) * h_2(n) = \sum_{k=0}^n \left(\frac{1}{3}\right)^k \left(\frac{1}{3}\right)^{n-k}$$

$$= \sum_{k=0}^n \left(\frac{1}{3}\right)^n$$

$$= (n+1) \left(\frac{1}{3}\right)^n u(n)$$

$$g(n) = h_1 * h_2 - h_1$$

$$= (n+1) \left(\frac{1}{3}\right)^n u(n) - \left(\frac{1}{2}\right)^n u(n)$$

(a) $h_3(n) = u(n)$

$$h(n) = h_3(n) * g(n)$$

$$= \sum_{k=0}^n (k+1) \left(\frac{1}{3}\right)^k - \left(\frac{1}{2}\right)^k$$

$$= \frac{1 - (n+2) \left(\frac{1}{3}\right)^{n+1} + (n+1) \left(\frac{1}{3}\right)^{n+2}}{(1 - \frac{1}{3})^2} - \frac{1 - \left(\frac{1}{2}\right)^{n+1}}{1 - \frac{1}{2}}$$

$$= \frac{9}{4} \left(1 - \frac{n+2}{3^{n+1}} + \frac{n+1}{3^{n+2}} \right) - 2 \left(1 - \left(\frac{1}{2}\right)^{n+1} \right)$$

$$= \left[\frac{1}{4} - \frac{2n+5}{4 \cdot 3^n} + \frac{1}{2^n} \right] u(n)$$

(b) $e(n) = h(n) * u(n)$

$$= \sum_{k=0}^n h(k)$$

$$= \sum_{k=0}^n \left(\frac{1}{4} + \frac{1}{2^k} - \frac{2k+5}{4 \cdot 3^k} \right)$$

$$= \frac{n+1}{4} + 2 \left(1 - \left(\frac{1}{2}\right)^{n+1} \right) - \left(\frac{9}{8} - \frac{3n+19}{8 \cdot 3^{n+1}} \right)$$

Multiple choice questions

Ans-6.1 (b)

Ans-6.2

$$z \rightarrow e^{j\omega}$$

(c)

Ans-6.3

$$x(n) = \delta(n-1)$$

$$h(n) = \delta(n+1)$$

$$x(n) * h(n) = \delta(n)$$

$$z \rightarrow 1$$

(b)

Ans-6.4 (a)

$$\text{Ans-6.5 } u(n) + u(n) = (n+1)u(n)$$

(a)

Ans-6.6

$$\delta(n-u) \xrightarrow{z_0 T_0} z^{-u}$$

(b)

Ans-6.7

$$2^n u(n) - 3^n u(n-1)$$

(c)

$$\text{Ans-6.8 } x(n) = 2^n u(n-1)$$

$$X(z) = \sum_{n=1}^{\infty} (2z)^{n-1}$$

$$= \frac{2z^{-1}}{1-2z^{-1}}$$

$$= \frac{2}{z-2}$$

$$|2z-1| < 1$$

$$\Rightarrow |z| > 2$$

Z transform

but

$$2^n u(n) - 2^n (u(n-1))$$

$$z \delta(n)$$

(c)

Ans-6.9 10

$$x_1(n) * x_2(n)$$

(c)

Ans-6.9

$$|z| > 2$$

$$e^{-2} < |z| < 2$$

$$|z| < e^{-2}$$

(c) 3

Ans-6.11

$$x(n) = \sum_{k=-\infty}^0 \delta(n-k) = u(-n) \quad X(z) = \frac{1}{1-z}$$

(c)

Ans-6.12

$$x_2(n) = x_1(-n)$$

(b)

Ans-6.13

$$y(n) - a y(n-1) = x(n)$$

$$Y(z) - az^{-1} Y(z) = X(z)$$

$$H(z) = \frac{1}{1-az^{-1}}$$

Pole at $z=a$

$$|z| < 1$$

$$\Rightarrow |a| < 1$$

(b)

Ans-6.14

$$X(z) = 0.6 + 0.2z^{-1} + 0.7z^{-2} + 0.5z^{-3}$$

$$ROC = \frac{0.6z^3 + 0.2z^2 + 0.7z + 0.5}{z^3}$$

$$|z| > 0$$

(a)

Ans-6.15

$$H(z)X(z) = \sum_{k=-\infty}^{\infty} h(k)x(n-k)$$

(c)

Ans-6.16

$$|z| > R \quad \left| \frac{z}{a} \right| > R$$

(a)

Ans-6.17

$$z^2 + 2z + 8 + 1z^{-1} + 6z^{-2}$$

(b)

Ans-6.18

(c)

PROBLEMS

Ans-6.1

$$x(n-2) - 9x(n-1) + 16x(n) = 0 \quad \text{--- (1)}$$

$$x(-1) = 1, \quad x(-2) = 9$$

$$z + \{x(n-1)\} = z^{-1}X(z) + z(-1)$$

$$z + z^{-1}X(z)$$

$$z + \{x(n-2)\} = z^{-2}X(z) + z^{-1} + 9z^{-2}X(z) + x(-1)z^{-1} + x(-2)$$

taking Z.T. of eqn (1)

$$z^{-2}X(z) + z^{-1} + 9 - 9z^{-1}X(z) - 9 + 16(X(z)) = 0$$

$$\therefore X(z) = \frac{-z^{-1}}{z^{-2} - 9z^{-1} + 16}$$

$$h_2(n) * h_3(n) = \delta(n) - \frac{1}{2} \delta(n-1)$$

$$h_2(2) * (h_3 + h_2) = -\frac{1}{2} \delta(n-1)$$

$$h_1(n) = \delta(n) + \frac{1}{2} \delta(n-1)$$

$$h(n) = \delta(n) + \frac{1}{2} \delta(n-1) + \left(-\frac{1}{2} \delta(n-1)\right)$$

$$= \delta(n)$$

Ans - 3.15

$$y_1(n) = x(n) * h_1(n)$$

$$y_2(n) = x(n) * h_2(n)$$

$$y_3(n) = -x(n)$$

$$y_s(n) = x(n) * h_1(n) + x(n) * h_2(n) - x(n)$$

$$x(n) = x(n) * \delta(n)$$

$$y_s(n) = x(n) * (h_1(n) + h_2(n) - \delta(n))$$

$$y_3(n) = y_s(n) * h_3(n)$$

$$= x(n) * (h_1(n) + h_2(n) - \delta(n)) * h_3(n)$$

$$y(n) = y_3(n) * h_1(n)$$

On comparing $y(n) = x(n) * h(n)$

$$\therefore h(n) = (h_1(n) + h_2(n) - \delta(n)) * h_3(n) * h_1(n)$$

Ans-603

$$x(n-n_0) = y(n)$$

$$(a) X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

$$Y(z) = \sum_{n=-\infty}^{\infty} x(n-n_0) z^{-n}$$

$$k = n - n_0$$

$$\Rightarrow n = k + n_0$$

$$= \sum_{k=-\infty}^{\infty} z^{-n_0} x(k) z^{-k}$$

$$= z^{-n_0} X(z)$$

$$(b) y(n) = g(n-2)$$

$$h(n) = g(n)$$

$$y(n) = x(n) * h(n)$$

$$Y(z) = X(z) H(z)$$

$$\Rightarrow z^{-2} = X(z) (1)$$

$$\Rightarrow X(z) = z^{-2}$$

$$\Rightarrow x(n) = \delta(n-2)$$

Ans-604

$$y(n) + 8y(n-1) + 2y(n-2) = 4$$

$$= x(n-1) + 8x(n-2)$$

$$y(0) = 1, y(1) = 2$$

$$x(n) = u(n)$$

$$x(n-1) = 1 \quad x(n) = 1 \text{ for } n \geq 2$$

$$x(n-2) = 1$$

$$y(n) + 8y(n-1) + 2y(n-2) = 4$$

$$\text{Let } y(n) + 8y(n-1) + 2y(n-2) = 0$$

$$y(n) = r^n$$

$$r^n + 8r^{n-1} + 2r^{n-2} = 0$$

$$\Rightarrow r^2 + 8r + 2 = 0$$

$$\Rightarrow (r+1)(r+2) = 0$$

$$r_1 = -1, r_2 = -2$$

$$y_h(n) = A(-1)^n + B(-2)^n$$

$$y_p(n) = C$$

$$C + 8C + 2C = 4$$

$$\Rightarrow C = \frac{2}{3}$$

$$y(n) = A(-1)^n + B(-2)^n + \frac{2}{3}$$

$$n = 0$$

$$\Rightarrow A + B + \frac{2}{3} = 4$$

$$\Rightarrow A + B = \frac{10}{3}$$

$$n = 1$$

$$A + 2B = -\frac{4}{3}$$

$$\Rightarrow A = 2 \text{ and } B = -\frac{5}{3}$$

$$y(n) = 2(-1)^n - \frac{5}{3}(-2)^n + \frac{2}{3}$$

Ans-605 To prove:

$$X(z) = X^*(z^*)$$

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

$$x^*(n) = x(n)$$

$$(a) \quad X^*(z^*) = \left[\sum_{n=-\infty}^{\infty} x(n) (z^*)^{-n} \right]^*$$

$$= \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

$$(b) \quad X(z) = X^*(z^*)$$

$$X(z_0) = 0$$

$$X^*(z_0^*) = 0$$

$$\text{Let } X(z_0^*) = 0$$

z_0^* is also zero

$$X(z) = X^*(z^*)$$

(c)

$$z = e^{j\theta}$$

$$z^* = e^{-j\theta}$$

$$z_2 = e^{-j\theta}$$

$$z_2 = \frac{1}{e^{j\theta}}$$

$$z = e^{j\theta} \text{ on unit circle}$$

$$z = e^{-j\theta}$$

Ans-606

$$x_2(n) = x_1(-n)$$

$$(a) \quad X_2(z) = X_1(z^{-1})$$

$$z = \frac{1}{z_0}$$

$$X_2(z) = \sum_{n=-\infty}^{\infty} x_1(-n) z^{-n}$$

$$k = -n$$

$$n = -k$$

$$= \sum_{k=-\infty}^{\infty} x_1(k) z^k$$

$$= \sum_{k=-\infty}^{\infty} x_1(k) (z^{-1})^{-k}$$

$$X_1(\omega) \sum_{k=-\infty}^{\infty} x_1(k) \omega^{-k}$$

$$X_2(z) = X_1(z^{-1})$$

$$= X_1\left(\frac{1}{z}\right)$$

$$(b) \quad z = z_0$$

$$X_2(z) = X_1\left(\frac{1}{z}\right)$$

$$\frac{1}{z} = z_0 \Rightarrow z = \frac{1}{z_0} \text{ is a pole or zero}$$

$$\text{of } X_2(z)$$

$$X(z) = \frac{-z}{(1-3z)(1-6z)}$$

$$1) X(z) = \frac{z-1}{(1-3z)(1-6z)}$$

$$= \frac{1}{3} \frac{1}{z-\frac{1}{3}} - \frac{1}{3} \frac{1}{z-\frac{1}{6}}$$

$$2) X(z) = \frac{1}{3} \left(\frac{1}{1-\frac{1}{3}z^{-1}} - \frac{1}{1-\frac{1}{6}z^{-1}} \right)$$

$$x(n) = \frac{1}{3} \left(\left(\frac{1}{3}\right)^n u(n) - \left(\frac{1}{6}\right)^n u(n-1) \right)$$

Ans-6.2

$$X(z) = \frac{1}{1-z^{-2}}$$

1. $|z| > 1$ - causal

2. $|z| < 1$ - non causal

$$(a) \frac{1}{1-z^{-2}}$$

$$|z^{-2}| < 1$$

$$1) |z| > 1$$

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

$$= 1 + z^{-2} + z^{-4} + \dots$$

$$x(n) = \delta(n) + \delta(n-2) + \delta(n-4) + \dots$$

$$z \left\{ \dots, 0, 0, \underset{\uparrow}{1}, 0, 1, 0, 1, \dots \right\}$$

(1)

$$(b) |z| < 1$$

$$X(z) = \frac{z^{-2}}{1-z^2}$$

$$= z^{-2} (1 + z^2 + z^4 + z^6 + \dots)$$

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

$$x(n) = -1, n = -2, -4, -6, \dots$$

$$z = \sum_{k=1}^{\infty} \delta(n+2k)$$

$$z \left\{ \dots, -1, 0, -1, 0, -1, 0, \dots \right\}$$

$$(b) \quad y(n) - 1.5y(n-1) + 0.5y(n-2) = 0$$

$$y(-1) = 1, \quad y(-2) = 0$$

$$y - 1.5(z^{-1}y + y(-1)) + 0.5(z^{-2}y + y(-1)z^{-1} + y(-2)) = 0$$

$$\Rightarrow Y(z) = \frac{1.5 - 0.5z^{-1}}{1 - 1.5z^{-1} + 0.5z^{-2}} \\ = \frac{1.5 - 0.5z^{-1}}{(1-z^{-1})(1-0.5z^{-1})}$$

$$Y(z) = \frac{A}{1-z^{-1}} + \frac{B}{1-0.5z^{-1}}$$

$$\Rightarrow A = 0.5, \quad B = 1$$

$$Y(z) = \frac{0.5}{1-z^{-1}} + \frac{1}{1-0.5z^{-1}}$$

$$\Rightarrow y(n) = \frac{1}{2} + \left(\frac{1}{2}\right)^n$$

$$(c) \quad y(n) = \frac{1}{2}y(n-1) + x(n)$$

$$x(n) = \left(\frac{1}{3}\right)^n u(n)$$

$$y(-1) = 1$$

$$Z\{a^n u(n)\} = \frac{1}{1-az^{-1}}$$

$$X(z) = \frac{1}{1-\frac{1}{3}z^{-1}}$$

$$Y(z) = \frac{1}{2}Z + \frac{1}{2}y(n-1) + X(z)$$

$$Z = \frac{1}{2}(z^{-1}Y(z) + y(-1)) + X(z)$$

$$\Rightarrow Y(z) = \frac{1}{2}z^{-1}Y + \frac{1}{2} + \frac{1}{1-\frac{1}{3}z^{-1}}$$

$$\Rightarrow Y = \frac{1}{2}\left(1 - \frac{1}{3}z^{-1}\right) + 1$$

$$\left(1 - \frac{1}{2}z^{-1}\right)\left(1 - \frac{1}{3}z^{-1}\right)$$

$$= \frac{3}{2} - \frac{1}{2}z^{-1}$$

$$\left(1 - \frac{1}{2}z^{-1}\right)\left(1 - \frac{1}{3}z^{-1}\right)$$

$$= \frac{7}{2} - \frac{2}{3}z^{-1}$$

$$\frac{1 - \frac{1}{2}z^{-1}}{1 - \frac{1}{2}z^{-1}} = \frac{1 - \frac{1}{3}z^{-1}}{1 - \frac{1}{3}z^{-1}}$$

$$\Rightarrow y(n) = \frac{7}{2}\left(\frac{1}{2}\right)^n - 2\left(\frac{1}{3}\right)^n$$

$$\left(\frac{-1-\sqrt{5}}{4}\right)^n$$

Ans-6.9

$$H(z) = \frac{1 - 2z^{-1} + z^{-2}}{(1 - z^{-1})(1 - 0.5z^{-1})(1 - 0.2z^{-1})}$$

(a) zeros

$$(1 - z^{-1})(1 - z^{-1} + z^{-2}) = 0$$

$$\Rightarrow z = \frac{1 \pm j\sqrt{3}}{2} = 1 \angle \pm 60^\circ$$

$$\& z = 1$$

$$(1 - z^{-1})(1 - 0.5z^{-1})(1 - 0.2z^{-1})$$

$$\Rightarrow p_1 = 1, p_2 = 0.5, p_3 = 0.2$$

$$\text{Poles: } 1, 0.5, 0.2$$

Hence 1 is cancelled

(b)

$$H(z) = \frac{1 - z^{-1} + z^{-2}}{(1 - 0.5z^{-1})(1 - 0.2z^{-1})}$$

$$= 1 + \frac{A}{1 - 0.5z^{-1}} + \frac{B}{1 - 0.2z^{-1}}$$

$$= 1 - \frac{2}{1 - 0.5z^{-1}} + \frac{2}{1 - 0.2z^{-1}}$$

$$\Rightarrow h(n) = \delta(n) - 2(0.5)^n u(n) + 2(0.2)^n u(n)$$

$$(c) |z| > 1$$

$$|z| > 0.5$$

Causal system is stable

Ans-6.10

$$y(n) = \sum_{k=-\infty}^n x(k)$$

$$y(n) = \sum_{k=-\infty}^{\infty} x(k) u(n-k)$$

$$= x(n) * u(n)$$

$$Z\{x(n) * u(n)\} = X(z) U(z)$$

$$U(z) = \frac{1}{1 - z^{-1}}$$

$$|z| > 1$$

$$Y(z) = \frac{X(z)}{1 - z^{-1}}$$

(b)

$$x(n) = (n+1)u(n)$$

$$Z\{n u(n)\} = \frac{z^{-1}}{(1 - z^{-1})^2}$$

$$(n+1)u(n) = n u(n) + u(n)$$

$$X(z) = \frac{z^{-1}}{(1 - z^{-1})^2} + \frac{1}{1 - z^{-1}}$$

$$= \frac{1}{(1 - z^{-1})^2}$$

$$= \frac{z^2}{(z-1)^2}$$

$$|z| > 1$$

Ans-6.7

$$h(n) = \begin{cases} 3^n, & n < 0 \\ \left(\frac{1}{2}\right)^n, & n \geq 0 \end{cases}$$

$$x(n) = u(n)$$

$$y(n) = x(n) * h(n)$$

$$= \sum_{k=-\infty}^n h(k)$$

$$h(k) = 3^k, \quad n < 0$$

$$y(n) = \sum_{k=-\infty}^n 3^k$$

$$= \frac{3^n}{1-\frac{1}{3}} = \frac{3^n}{\frac{2}{3}} = \frac{3^{n+1}}{2}$$

$$n \geq 0$$

$$y(n) = \sum_{k=-\infty}^{-1} 3^k + \sum_{k=0}^n \left(\frac{1}{2}\right)^k$$

$$= \frac{3^{-1}}{1-\frac{1}{3}} + \frac{1 - \left(\frac{1}{2}\right)^{n+1}}{1 - \frac{1}{2}}$$

$$= \frac{1}{2} + 2 - \left(\frac{1}{2}\right)^n$$

$$y(n) = \frac{5}{2} - \left(\frac{1}{2}\right)^n$$

$$y(n) = \begin{cases} \frac{3^{n+1}}{2}, & n < 0 \\ \frac{5}{2} - 2^{-n}, & n \geq 0 \end{cases}$$

Ans-6.8

(a)

$$y(n) + \frac{1}{2}y(n-1) - \frac{1}{4}y(n-2) = 0$$

$$y(-1) = y(-2) = 1$$

$$z + \frac{1}{2}y(n-1) = z^{-1}Y(z) + y(-1)$$

$$z + \frac{1}{2}y(n-2) = z^{-2}Y(z) + z^{-1} + 1$$

$$Y + \frac{1}{2}(z^{-1}Y + 1) - \frac{1}{4}(z^{-2}Y + z^{-1} + 1) = 0$$

$$\Rightarrow Y(z) = \frac{\frac{1}{4}(z^{-1} - 1)}{1 + \frac{1}{2}z^{-1} - \frac{1}{4}z^{-2}}$$

$$\pi^2 + \frac{1}{2}\pi - \frac{1}{4}z = 0$$

$$\Rightarrow \pi = \frac{-1 \pm \sqrt{5}}{4}$$

$$y(n) = A\pi_1^n + B\pi_2^n$$

$$At n=0$$

$$y(0) = -\frac{1}{4} = A + B$$

$$At n=1$$

$$y(1) = \frac{3}{8} = A\pi_1 + B\pi_2$$

$$A = \frac{\sqrt{5}-1}{8}, \quad B = \frac{-\sqrt{5}+1}{8}$$

$$y(n) = \frac{\sqrt{5}-1}{8} \left(\frac{-1+\sqrt{5}}{4}\right)^n - \frac{\sqrt{5}+1}{8} \left(\frac{-1-\sqrt{5}}{4}\right)^n$$

Ans-6.13

$$y(n) = \sum_{k=0}^{\infty} x(n-kN)$$

$$0 \leq n \leq N-1$$

$$Y(z) = \sum_{n=-\infty}^{\infty} \sum_{k=0}^{\infty} x(n-kN) z^{-n}$$

$$n = m + kN$$

$$n = m + kN$$

$$= \sum_{k=0}^{\infty} z^{-kN} \sum_{m=0}^{\infty} x(m) z^{-m}$$

$$= X(z) \sum_{k=0}^{\infty} z^{-kN}$$

$$= \frac{X(z)}{1 - z^{-N}}$$

$$|z^{-N}| < 1$$

$$\Rightarrow |z| > 1$$

Ans-6.14

$$x(n) = u(n) + \left(-\frac{1}{2}\right)^n u(n)$$

$$y(n) = 6\left(-\frac{1}{4}\right)^n u(n) - 6\left(-\frac{1}{3}\right)^n u(n)$$

$$X(z) = \frac{1}{1-z^{-1}} + \frac{1}{1+\frac{1}{2}z^{-1}}$$

$$= \frac{2 - \frac{1}{2}z^{-1}}{(1-z^{-1})(1+\frac{1}{2}z^{-1})}$$

$$Y(z) = \frac{6}{1+\frac{1}{4}z^{-1}} - \frac{6}{1+\frac{1}{3}z^{-1}}$$

$$= \frac{\frac{1}{2}z^{-1}}{(1+\frac{1}{4}z^{-1})(1+\frac{1}{3}z^{-1})}$$

$$\frac{\frac{1}{2}z^{-1}}{(1+\frac{1}{4}z^{-1})(1+\frac{1}{3}z^{-1})}$$

$$H(z) = \frac{Y(z)}{X(z)}$$

$$= \frac{\frac{1}{2}z^{-1}}{(1+\frac{1}{4}z^{-1})(1+\frac{1}{3}z^{-1})} \frac{(1-z^{-1})(1+\frac{1}{2}z^{-1})}{2 - \frac{1}{2}z^{-1}}$$

$$= \frac{\frac{1}{4}z^{-1}(1-z^{-1})(1+\frac{1}{2}z^{-1})}{(1+\frac{1}{4}z^{-1})(1+\frac{1}{3}z^{-1})(1-\frac{1}{4}z^{-1})}$$

$$(b) (1-z^{-1}) \left(1+\frac{1}{2}z^{-1}\right)$$

$$= 1 - \frac{1}{2}z^{-1} - \frac{1}{2}z^{-2}$$

$$B(z) = \frac{1}{4}z^{-1} - \frac{1}{8}z^{-2} - \frac{1}{8}z^{-3}$$

$$\left(1+\frac{1}{4}z^{-1}\right)\left(1-\frac{1}{4}z^{-1}\right) = 1 - \frac{1}{16}z^{-2}$$

$$1 + \frac{1}{3}z^{-1}$$

$$A(z) = 1 + \frac{1}{3}z^{-1} - \frac{1}{16}z^{-2} - \frac{1}{48}z^{-3}$$

$$A(z)Y(z) = B(z)X(z)$$

$$y(n) + \frac{1}{3}y(n-1) - \frac{1}{16}y(n-2) - \frac{1}{48}y(n-3)$$

$$= \frac{1}{4}x(n-1) - \frac{1}{8}x(n-2) - \frac{1}{8}x(n-3)$$

Ans-6.15

$$X(z) = \frac{z}{(z-0.5)(z-1.5)}, 0.5 < |z| < 1.5$$

$$Y(z) = \frac{0.25z}{(z-0.25)(z-0.5)}, |z| > 0.5$$

$$X(z) = \frac{-0.5}{z-0.5} + \frac{1.5}{z-1.5}$$

$$Y(z) = \frac{0.25}{z-0.25} + \frac{0.5}{z-0.5}$$

$$aX(z) + b(Y(z))$$

$$-0.5a + 0.5b$$

$$\rightarrow a=b$$

$$z=0.25, 1.5$$

$$0.25 < |z| < 1.5$$

Largest occurs for $a=b$

Ans-6.16

$$H(z) = \frac{1+z^{-1}}{(1-az^{-1})(1-bz^{-1})}$$

$$z=a, z=b$$

(a) causal

$$|z| > \max(|a|, |b|)$$

(b) Non-causal

$$|z| < \min(|a|, |b|)$$

Causality depends on ROC,

not merely on a, b

Ans-6.17

$$|z| > 3, |z| < \frac{1}{2}, \frac{1}{2} < |z| < 3$$

$$z = \frac{1}{2}, z=3 \text{ poles}$$

$$2\left(z - \frac{1}{2}\right)(z-3)$$

$$2z^2 - 7z + 3$$

$$X(z) = \frac{z+2}{2\left(z - \frac{1}{2}\right)(z-3)}$$

$$\frac{z - \frac{1}{2}}{z - \frac{1}{2}} + \frac{1}{z-3}$$

$$\frac{1}{z-a} \xrightarrow{z^{-1}} a^{n-1}u(n-1)$$

$$\frac{1}{z-a} \xrightarrow{z^{-1}} -a^{n-1}u(-n)$$

$$(a) |z| > 3$$

$$x(n) = -\frac{1}{2}\left(\frac{1}{2}\right)^{n-1}u(n-1) + 3^{n-1}u(n-1)$$

$$\frac{1}{2}\left(\frac{1}{2}\right)^{n-1} = \left(\frac{1}{2}\right)^n$$

$$x(n) = \left(3^{n-1} - \left(\frac{1}{2}\right)^n\right)u(n-1)$$

$$(b) |z| < \frac{1}{2}$$

$$-\frac{1}{2} \frac{1}{z - \frac{1}{2}} \rightarrow \left(\frac{1}{2}\right)\left(\frac{1}{2}\right)^{n-1}u(-n)$$

$$= \left(\frac{1}{2}\right)^n u(-n)$$

$$\frac{1}{z-3} \rightarrow -3^{n-1}u(-n)$$

$$x(n) = \left(\left(\frac{1}{2}\right)^n - (3)^{n-1}\right)u(-n)$$

Ans-6.11

$$X(z) = \frac{1+2z^{-1}}{1-2z^{-1}+z^{-2}}$$

$$= \frac{1+2z^{-1}}{(1-z^{-1})^2}$$

(a) $x(n]$ causal

$$\frac{1}{(1-z^{-1})^2} = 1+2z^{-1}+3z^{-2}+\dots$$

$$X(z) = (1+2z^{-1})(1+2z^{-1}+3z^{-2}+\dots)$$

$$= 1+4z^{-1}+7z^{-2}+10z^{-3}+\dots$$

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

$$x(0) = 1$$

$$x(1) = 4$$

$$x(2) = 7 \dots$$

$$x(n) = (n+1)u(n)$$

$$ROC: |z| > 1$$

(b) $|z| < 1$

$$X(z) = \frac{z(z+2)}{(z-1)^2}$$

$$= (2z+z^2)(1+2z+3z^2+\dots)$$

$$= 2z+5z^2+8z^3+11z^4+\dots$$

$$z^n = z^{-(-n)}$$

$$x(-1) = 2, x(-2) = 5, x(-3) = 8, \dots$$

$$x(n) = (3n-1)u(-n-1)$$

Ans-6.12

(a) $x(n] = (8)^n u(-n-1)$

$$a^n u(-n-1) \xrightarrow{z^{-1}} \frac{-z}{z-a} \quad |z| < |a|$$

$$a = -3$$

$$X(z) = \frac{-z}{z+3}$$

$$z^{-1} \frac{1}{1+3z^{-1}} \quad |z| < 3$$

(b) $x(n] = \begin{cases} 1, & -5 \leq n \leq 5 \\ 0, & \text{otherwise} \end{cases}$

$$X(z) = \sum_{n=-5}^5 z^{-n}$$

$$= z^5 + z^4 + \dots + z^{-4} + z^{-5}$$

$$= z^5 \frac{1-z^{-11}}{1-z^{-1}} \quad 1 < |z| < 3$$

$$0 < |z| < \infty$$

(c) $x(n] = \begin{cases} (\frac{1}{3})^n, & n \geq 0 \\ 8^n, & n < 0 \end{cases}$

$$z(\frac{1}{3})^n u(n) + 8^n u(-n-1)$$

$$X_1(z) = \frac{1}{1-\frac{1}{3}z^{-1}} \quad |z| > \frac{1}{3}$$

$$X(z) = \frac{1}{1-\frac{1}{3}z^{-1}} - \frac{1}{1-8z^{-1}}$$

$$|z| > \frac{1}{3}$$

$$|z| < 3$$

$$\frac{1}{3} < |z| < 3$$

11. State whether the following statements are True/False

- | | | | |
|--------------|--------------|---------------|---------------|
| Ans-1. True | Ans-5. False | Ans-9. False | Ans-13. True |
| Ans-2. True | Ans-6. True | Ans-10. False | Ans-14. True |
| Ans-3. False | Ans-7. True | Ans-11. False | Ans-15. False |
| Ans-4. True | Ans-8. False | Ans-12. True | Ans-16. True |

111. Choose the right answer for following questions.

- | | | | | |
|------------|-------------|-------------|-------------|--------------------------|
| Ans-1. (c) | Ans-8. (b) | Ans-15. (b) | Ans-22. (a) | Ans-29. (d) |
| Ans-2. (a) | Ans-9. (a) | Ans-16. (b) | Ans-23. (a) | Ans-30. (c) Ans -36. (b) |
| Ans-3. (b) | Ans-10. (c) | Ans-17. (c) | Ans-24. (a) | Ans-31. (d) Ans -37. (b) |
| Ans-4. (d) | Ans-11. (b) | Ans-18. (c) | Ans-25. (d) | Ans-32. (a) |
| Ans-5. (c) | Ans-12. (d) | Ans-19. (a) | Ans-26. (d) | Ans-33. (d) |
| Ans-6. (b) | Ans-13. (c) | Ans-20. (a) | Ans-27. (b) | Ans-34. (a) |
| Ans-7. (d) | Ans-14. (b) | Ans-21. (d) | Ans-28. (a) | Ans-35. (b) |

IV. Answer the following question.

Ans-1. Two sided Z-transform

For a discrete time sequence $x(n)$,

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

One sided Z-transform

$$X^+(z) = \sum_{n=0}^{\infty} x(n) z^{-n}$$

For $n \geq 0$

Ans-2. The region of convergence is the set of values of z for which the Z-transform summation converges to a finite value

$$ROC = \{ z : X(z) \text{ is finite} \}$$

Ans-3. $x(\infty) = \lim_{z \rightarrow 1} (1-z^{-1}) X(z)$ (Final value theorem)

$x(\infty) = \lim_{z \rightarrow 1} \frac{z-1}{1} X(z)$ provided the stability conditions are satisfied.

Ans-4. $x(0) = \lim_{z \rightarrow \infty} X(z)$ For a causal sequence

$X(z) = \frac{z}{z-a}$

$x(0) = \lim_{z \rightarrow \infty} \frac{z}{z-a} = 1$

$x(0) = 1$

Ans-5. $u(n) = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$

$X(z) = \sum_{n=0}^{\infty} z^{-n} = \frac{1}{1-z^{-1}} = \frac{z}{z-1} \quad |z| > 1$

Ans-6. (i) Power series expansion method

(ii) Partial fraction expansion method

(iii) Contour-integration method

(iv) Long division method

Ans-7. The z transform function can be obtained from s domain transfer function.

$h(n) = h_c(nT)$

$H(z) = Z\{h_c(nT)\}$

$z = e^{sT}$

$$(c) \frac{1}{2} < |z| < 3$$

$$x(n) = -\frac{1}{2} \left(\frac{1}{2} \right)^{n-1} u(n-1) - 3^{n-1} u(-n)$$

$$= -\left(\frac{1}{2} \right)^n u(n-1) - 3^{n-1} u(-n)$$

Z-transform

Exercises

1. Fill in the blanks with appropriate words

Ans-1. region of convergence

Ans-2. $\log(1/5) \ln 2$

Ans-3. left half, interior

Ans-4. transfer function

Ans-5. left half, interior

Ans-6. right half, exterior

Ans-7. unit circle

Ans-8. impulse variant

Ans-9. aliasing

Ans-10. outside, largest

Ans-11. poles

Ans-12. anticausal sequences, $n \leq 0$

Ans-13. impulse response

Ans-14. $X^*(z^*)$

Ans-15. multiplying z^2

Ans-16. direct form-II

Ans-17. canonic structure

Ans-18. $M+N+1$

Ans-19. N

Ans-20. $N-1$

Ans-21. Lenné phase

Ans-12. $z = e^{sT}$

$$s = \sigma + j\omega$$

$$z = e^{\sigma T} \cdot e^{j\omega T}$$

$$e^{j(\omega + 2\pi/T)T} = e^{j\omega T}$$

$$s = \sigma + j\omega = \sigma + j\left(\omega + \frac{2\pi}{T}\right)$$

Hence, the mapping is many to one, not one to one

Ans-13. $y(n) = x(n) * h(n)$

Convolution $Y(z) = X(z)H(z)$

thus, convolution in the time domain becomes multiplication in the Z domain

Deconvolution

$$Y(z) = X(z)H(z)$$

$$X(z) = \frac{Y(z)}{H(z)}$$

thus, deconvolution in the time domain corresponds to division in the Z domain.

Ans-14. For an LTI discrete time system to be BIBO stable, its impulse response must be summable;

$$\sum_{n=-\infty}^{\infty} |h(n)| < \infty$$

The ROC must include unit circle

For a rational system, all poles lie inside the unit circle for a causal system.

Ans-8. For a discrete LTI system, the transfer function is defined as the ratio of Z transform of the output to input, assuming 0 initial conditions

$$H(z) = \frac{Y(z)}{X(z)} = Z\{h(n)\}$$

Ans-9.
$$H(z) = \frac{b_0 + b_1 z^{-1} + \dots + b_M z^{-M}}{1 + a_1 z^{-1} + \dots + a_N z^{-N}}$$

For N^{th} order system, the denominator contains terms up to z^{-N} .

$$H(z) = \frac{b_0 z^N + b_1 z^{N-1} + \dots + b_M z^{N-M}}{z^N + a_1 z^{N-1} + \dots + a_N}$$

Ans-10. Impulse invariant transformation is a technique for converting a continuous time system into a discrete time system such that the discrete time impulse response is obtained by sampling continuous time impulse response

$$h(n) = h_c(nT)$$

$$z = e^{sT}$$

Ans-11.

$$s = \sigma + j\omega$$

$$z = e^{sT}$$

$$z = e^{(\sigma + j\omega)T}$$

$$|z| = e^{\sigma T}$$

$$Lz = \omega T$$

the left half of s-plane maps inside the $\odot$
the $j\omega$ -axis maps onto unit circle.

$$x(n-n_0) = a^{n-n_0} u(n-n_0)$$

$$z^{-n_0} X(z)$$

$$x(n-n_0) \longleftrightarrow z^{-n_0} X(z)$$

Ans-18.

1. ROC does not contain any poles.
2. ROC is a ring or circular region centered at origin.
3. For a stable system, the ROC must include the unit circle.
4. ROC cannot contain a pole.
5. For a two sided sequence

$$|p_1| < |z| < |p_2|$$

Ans-15. $y(n) = f[x(n), x(n-1), x(n-2), \dots]$

An LTI discrete-time system is causal if its output at the present time depends only on the present and past input values.

and it must not depend on $x(n+1), x(n+2), \dots$

For an LTI system:

$$h(n) = 0, n < 0$$

ROC: $|z| > \text{magnitude of outermost pole}$

Ans-16.

1. For finite-duration sequence

$$x(n) = 0 \text{ except for finite values of } n$$

2. Right sided infinite sequence

$$x(n) = a^n u(n)$$

$$X(z) = \frac{z}{z-a}$$

$$\text{ROC: } |z| > |a|$$

3. For left sided sequence

$$\text{ROC: } |z| < |a|$$

4. Two sided sequence

$$|a_1| < |z| < |a_2|$$

Ans-17.

$$x(n) \xrightarrow{z} X(z)$$

$$x(n-n_0) \xrightarrow{z} z^{-n_0} X(z)$$

For e.g. $x(n) = a^n u(n)$

$$X(z) = \frac{z}{z-a}$$